A Technical Introduction to Variational autoencoder

$$\mathcal{L} = \mathbb{E}_{q_\phi}[\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z))$$

Section 1

the usual reconstruction error part which seeks to ensure that the encoder-then-decoder mapping x ■ D θ (E ψ (x)) {\displaystyle x\mapsto D_{\theta }(E_{\psi }(x))} is as close to the identity map as possible; the sampling is done at run time from the empirical distribution P r e a l {\displaystyle \mathbb {P} ^{real}} of objects available (e.g., for MNIST or IMAGENET this will be the empirical probability law of all images in the dataset). This gives the term: E x ~ P r e a l [■ x − D θ (E φ (x)) ■ 2 2] {\displaystyle \mathbb {E} _{x\sim \mathbb {P} ^{real}}\left[\|x-D_{\theta }(E_{\phi }(x))\|_{2}^{2}\right]} . a variational part that ensures that, when the empirical distribution P r e a l {\displaystyle \mathbb {P} ^{real}} is passed through the encoder E φ {\displaystyle E_{\phi }} , we recover the target distribution, denoted here μ (d z) {\displaystyle \mu (dz)} that is usually taken to be a Multivariate normal distribution. We will denote E φ ■ P r e a l {\displaystyle E_{\phi }\sharp \mathbb {P} ^{real}} this pushforward measure which in practice is just the empirical distribution obtained by passing all dataset objects through the encoder E φ {\displaystyle E_{\phi }} . In order to make sure that E φ ■ P r e a l {\displaystyle E_{\phi }\sharp \mathbb {P} ^{real}} is close to the target μ (d z) {\displaystyle \mu (dz)} , a Statistical distance d {\displaystyle d} is invoked and the term d (μ (d z) , E φ ■ P r e a l) 2 {\displaystyle d\left(\mu (dz),E_{\phi }\sharp \mathbb {P} ^{real}\right)^{2}} is added to the loss. We obtain the final formula for the loss:

Section 2

In machine learning, a variational autoencoder (VAE) is an artificial neural network architecture introduced by Diederik P. Kingma and Max Welling in 2013. It is part of the families of probabilistic graphical models and variational Bayesian methods. In addition to being seen as an autoencoder neural network architecture, variational autoencoders can also be studied within the mathematical formulation of variational Bayesian methods, connecting a neural encoder network to its decoder through a probabilistic latent space (for example, as a multivariate Gaussian distribution) that corresponds to the parameters of a variational distribution. Thus, the encoder maps each point (such as an image) from a large complex dataset into a distribution within the latent space, rather than to a single point in that space. The decoder has the opposite function, which is to map from the latent space to the input space, again according to a distribution (although in practice, noise is rarely added during the decoding stage). By mapping a point to a distribution instead of a single point, the network can avoid overfitting the training data. Both networks are typically trained together with the usage of the reparameterization trick, although the variance of the noise model can be learned separately. Although this type of model was initially designed for unsupervised learning, its effectiveness has been proven for semi-supervised learning and supervised learning.

Section 3

the sliced Wasserstein distance used by S Kolouri, et al. in their VAE the energy distance implemented in the Radon Sobolev Variational Auto-Encoder the Maximum Mean Discrepancy distance used in the MMD-VAE the Wasserstein distance used in the WAEs kernel-based distances used in the Kernelized Variational Autoencoder (K-VAE)

Section 4

To efficiently search for θ ∗ , φ ∗ = argmax θ , φ L θ , φ (x) {\displaystyle \theta ^{*},\phi ^{*}={\underset {\theta ,\phi }{\operatorname {argmax} }}\,L_{\theta ,\phi }(x)} the typical method is gradient ascent. It is straightforward to find ∇ θ E z ~ q φ (· | x) [ln ■ p θ (x , z) q φ (z | x)] = E z ~ q φ (· | x) [∇ θ ln ■ p θ (x , z) q φ (z | x)] {\displaystyle \nabla _{\theta }\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln {\frac {p_{\theta }(x,z)}{q_{\phi }({z|x})}}\right]=\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\nabla _{\theta }\ln {\frac {p_{\theta }(x,z)}{q_{\phi }({z|x})}}\right]} However, ∇ φ E z ~ q φ (· | x) [ln ■ p θ (x , z) q φ (z | x)] {\displaystyle \nabla _{\phi }\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln {\frac {p_{\theta }(x,z)}{q_{\phi }({z|x})}}\right]} does not allow one to put the ∇ φ {\displaystyle \nabla _{\phi }} inside the expectation, since φ {\displaystyle \phi } appears in the probability distribution itself. The reparameterization trick (also known as stochastic backpropagation) bypasses this difficulty. The most important example is when z ~ q φ (· | x) {\displaystyle z\sim q_{\phi }(\cdot |x)} is normally distributed, as N (μ φ (x) , Σ φ (x)) {\displaystyle {\mathcal {N}}(\mu _{\phi }(x),\Sigma _{\phi }(x))} .

Section 5

D K L (q φ (z | x) ■ p θ (z | x)) = E z ~ q φ (· | x) [ln ■ q φ (z | x) p θ (z | x)] = E z ~ q φ (· | x) [ln ■ q φ (z | x) p θ (x) p θ (x , z)] = ln ■ p θ (x) + E z ~ q φ (· | x) [ln ■ q φ (z | x) p θ (x , z)] . {\displaystyle {\begin{aligned}D_{KL}(q_{\phi }({z|x})\parallel p_{\theta }({z|x}))&=\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln {\frac {q_{\phi }(z|x)}{p_{\theta }(z|x)}}\right]\\&=\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln {\frac {q_{\phi }({z|x})p_{\theta }(x)}{p_{\theta }(x,z)}}\right]\\&=\ln p_{\theta }(x)+\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln {\frac {q_{\phi }({z|x})}{p_{\theta }(x,z)}}\right].\end{aligned}}}

Section 6

Now, define the evidence lower bound (ELBO): L θ , φ (x) := E z ~ q φ (· | x) [ln ■ p θ (x , z) q φ (z | x)] = ln ■ p θ (x) − D K L (q φ (· | x) ■ p θ (· | x)) {\displaystyle L_{\theta ,\phi }(x):=\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln {\frac {p_{\theta }(x,z)}{q_{\phi }({z|x})}}\right]=\ln p_{\theta }(x)-D_{KL}(q_{\phi }({\cdot |x})\parallel p_{\theta }({\cdot |x}))} Maximizing the ELBO θ ∗ , φ ∗ = argmax θ , φ L θ , φ (x) {\displaystyle \theta ^{*},\phi ^{*}={\underset {\theta ,\phi }{\operatorname {argmax} }}\,L_{\theta ,\phi }(x)} is equivalent to simultaneously maximizing ln ■ p θ (x) {\displaystyle \ln p_{\theta }(x)} and minimizing D K L (q φ (z | x) ■ p θ (z | x)) {\displaystyle D_{KL}(q_{\phi }({z|x})\parallel p_{\theta }({z|x}))} . That is, maximizing the log-likelihood of the observed data, and minimizing the divergence from the approximate posterior q φ (· | x) {\displaystyle q_{\phi }(\cdot |x)} to the exact posterior p θ (· | x) {\displaystyle p_{\theta }(\cdot |x)} . The form given is not very convenient for maximization, but the following, equivalent form, is: L θ , φ (x) = E z ~ q φ (· | x) [ln ■ p θ (x | z)] − D K L (q φ (· | x) ■ p θ (·)) {\displaystyle L_{\theta ,\phi }(x)=\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\ln p_{\theta }(x|z)\right]-D_{KL}(q_{\phi }({\cdot |x})\parallel p_{\theta }(\cdot))} where ln ■ p θ (x | z) {\displaystyle \ln p_{\theta }(x|z)} is implemented as − 1 2 ■ x − D θ (z) ■ 2 2 {\displaystyle -{\frac {1}{2}}\|x-D_{\theta }(z)\|_{2}^{2}} , since that is, up to an additive constant, what x | z ~ N (D θ (z) , I) {\displaystyle x|z\sim {\mathcal {N}}(D_{\theta }(z),I)} yields. That is, we model the distribution of x {\displaystyle x} conditional on z {\displaystyle z} to be a Gaussian distribution centered on D θ (z) {\displaystyle D_{\theta }(z)} . The distribution of q φ (z | x) {\displaystyle q_{\phi }(z|x)} and p θ (z) {\displaystyle p_{\theta }(z)} are often also chosen to be Gaussians as z | x ~ N (E φ (x) , σ φ (x) 2 I) {\displaystyle z|x\sim {\mathcal {N}}(E_{\phi }(x),\sigma _{\phi }(x)^{2}I)} and z ~ N (0 , I) {\displaystyle z\sim {\mathcal {N}}(0,I)} , with which we obtain by the formula for KL divergence of Gaussians: L θ , φ (x) = − 1 2 E z ~ q φ (· | x) [■ x − D θ (z) ■ 2 2] − 1 2 (N σ φ (x) 2 + ■ E φ (x) ■ 2 2 − 2 N ln ■ σ φ (x)) + C o n s t {\displaystyle L_{\theta ,\phi }(x)=-{\frac {1}{2}}\mathbb {E} _{z\sim q_{\phi }(\cdot |x)}\left[\|x-D_{\theta }(z)\|_{2}^{2}\right]-{\frac {1}{2}}\left(N\sigma _{\phi }(x)^{2}+\|E_{\phi }(x)\|_{2}^{2}-2N\ln \sigma _{\phi }(x)\right)+Const} Here N {\displaystyle N} is the dimension of z {\displaystyle z} . For a more detailed derivation and more interpretations of ELBO and its maximization, see its main page.